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Investigation of Cotton Stalk-Derived Hydrothermal Bio-Oil: Effects of Mineral Acid/Base and Oxide Additions

Libo Zhang [1,2,3,*,†], Jianing Wang [2,†], Hui Ming [3], Hanjun Hu [1], Xintong Dou [3], Yepeng Xiao [2], Lihua Cheng [2] and Zhun Hu [1,*]

[1] School of Chemical Engineering and Technology, Xi'an Jiaotong University, Xi'an 710049, China; huhanjun@peacockedu.com
[2] Guangdong Provincial Engineering & Technology Center for Corrosion and Safety in Petrochemical Industry, School of Chemical Engineering, Guangdong University of Petrochemical Technology, Maoming 525000, China; 2022216840@st.cupk.edu.cn (J.W.); yepengxiao@gdupt.edu.cn (Y.X.); chenlh@gdupt.edu.cn (L.C.)
[3] State Key Laboratory of Heavy Oil Processing, College of Engineering, China University of Petroleum-Beijing at Karamay, Karamay 834000, China; mh@cupk.edu.com (H.M.); douxintong@student.cup.edu.cn (X.D.)
* Correspondence: zhanglibo@cupk.edu.cn (L.Z.); huzhun@mail.xjtu.edu.cn (Z.H.)
† Those authors contribute equally to this work.

Abstract: Hydrothermal liquefaction technology (HTL) is a promising thermochemical method to convert biomass into novel liquid fuels. The introduction of oxides and inorganic acids/bases during the hydrothermal process significantly impacts the yield and composition of bio-oil. However, systematic research on their effects, especially at lower temperatures, remains limited. In this paper, we examine the effects of acidity and alkalinity on cotton stalk hydrothermal bio-oil by introducing homogeneous acids and bases. Given the operational challenges associated with product separation using homogeneous acids and bases, this paper further delves into the influence of heterogeneous oxide catalysts (possessing varying degrees of acidity and alkalinity, as well as distinct microstructures and pore architectures) on the production of cotton stalk hydrothermal bio-oil. The effects of nanoscale oxides (CeO_2, TiO_2, ZnO, Al_2O_3, MgO and SiO_2) and homogeneous acid–base catalysts (NaOH, K_2CO_3, Na_2CO_3, KOH, HCl, H_2SO_4, HNO_3) on the quality of cotton stalk bio-oil under moderate hydrothermal conditions (220 °C, 4 h) were investigated. Characterization techniques including infrared spectroscopy, thermogravimetric analysis, elemental analysis, and GC-MS were employed. The results revealed that CeO_2 and NaOH achieved the highest bio-oil yield due to Ce^{3+}/Ce^{4+} redox reactions, OH-LCC disruption, and ionic swelling effects. Nano-oxides enhanced the formation of compounds like N-ethyl formamide and aliphatic aldehydes while suppressing nitrogen-containing aromatics. The total pore volume and average pore width of oxides negatively correlated with their catalytic efficiency. CeO_2 with low pore volume and width exhibited the highest energy recovery. The energy recovery of cotton stalk bio-oil was influenced by both acid and base sites on the oxide surface, with a higher weak base content favoring higher yields and a higher weak acid content inhibiting them. The findings of this research are expected to provide valuable insights into the energy utilization of agricultural solid waste, such as cotton stalks, as well as to inform the design and development of highly efficient catalysts.

Keywords: nanometer oxide; catalytic hydrothermal liquefaction; cotton stalk; bio-oil; energy recovery rate

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1. Introduction

Since the dawn of the industrial revolution, fossil fuels, predominantly coal, oil, and natural gas, have served as the cornerstone of global energy development. Nevertheless, the excessive extraction of these fuels has given rise to concerns over resource depletion and environmental pollution. Consequently, governments worldwide have been actively

promoting the utilization of renewable energy sources as viable alternatives to fossil fuels [1]. Among renewable energy sources, including solar energy, wind energy, and biomass energy, biomass energy stands out as one of the most promising renewable resources for mitigating the future use of fossil fuels. This is attributed to its renewable nature, ubiquity, and environmental friendliness, making it a viable alternative to fossil fuels [2]. Globally, lignocellulosic biomass is abundantly available and primarily comprises cellulose, hemicellulose, and lignin. This biomass material, renowned for its vast quantity, low cost, and rapid growth rate, holds immense potential for diverse applications, making it a highly attractive renewable resource [3]. Cotton stalks, generated in large quantities during cotton cultivation, are a crucial biomass resource. Fully utilizing these abundant discarded cotton stalks from local fields holds significant potential for catalytically converting solid waste into energy sources, particularly in terms of harnessing all biomass components. This approach not only promotes the utilization of agricultural and forestry waste for biomass resourcefulness but also provides technological support for the economic development of local regions.

Common conversion methods for biomass raw materials encompass physicochemical, thermochemical, and biological approaches. Amongst these, hydrothermal conversion technology emerges as a pivotal thermochemical method. Its distinct advantage lies in the elimination of the prerequisite for preheating and drying the raw materials, significantly reducing energy consumption. Furthermore, its relatively moderate reaction conditions facilitate the production of a more stable hydrothermal bio-oil, rendering it a more versatile alternative compared to other thermochemical conversion methods, such as pyrolysis. This broadened range of applicability underscores the significant potential of hydrothermal conversion in biomass utilization, making it a promising technology for sustainable energy production [4–7]. In the process of hydrothermal conversion, water not only serves as a solvent but also acts as one of the reactants. Under the high-temperature and high-pressure conditions employed in the hydrothermal process, the properties of water undergo significant changes. These include a sharp decrease in dielectric constant, reduced viscosity, increased diffusion coefficient, decreased ion product, enhanced non-polarity, and even miscibility with oil [8]. Under the influence of water, macromolecular carbohydrates in biomass raw materials can be degraded into monomeric carbohydrates at temperatures above 180 °C [9]. Through a series of reactions including hydrolysis, decarboxylation, deamination, re-polymerization, and Maillard reactions, these carbohydrates ultimately yield bio-oil, water-soluble products, gases, and solids. Due to the vast array of components in biomass and their complex interactions, the precise reaction mechanisms remain elusive [10].

The bio-oil derived from hydrothermal liquefaction, despite its potential as a liquid fuel and chemical raw material due to its ease of transportation and high energy density, faces challenges in its development and application due to limited production and quality [11]. To address these issues, researchers often employ both homogeneous and heterogeneous catalysts to enhance the quality and yield of the bio-oil. However, homogeneous catalysts, while effective, are difficult to recover, leading to increased costs and environmental concerns. Conversely, heterogeneous catalysts offer advantages in terms of recovery efficiency, thermal stability, and equipment corrosion resistance. Research indicates that specific metals and metal oxides, such as Pt, Pd, Ru, Ni, Co, MgO, Al_2O_3, and others, can significantly improve bio-oil quality through various catalytic reactions [12,13]. Nevertheless, there is a need for further investigation into the detailed impacts of oxide structures, surface acidity, and alkalinity of bio-oils, especially at low temperatures and with respect to lignocellulosic materials. The current literature primarily focuses on oxides as supports for catalysts, leaving a gap in our understanding of the catalytic properties of oxides themselves [14–17].

Despite the preponderance of existing research emphasizing the pivotal role of acid-alkalinity in the hydrothermal transformation of lignocellulosic biomass, a comprehensive, systematic comparative analysis of homogeneous acid–base systems versus solid oxide

acid–base catalysts has yet to be undertaken. Notably, the intricacies of the hydrothermal conversion process are further compounded by the complex interplay between the pore architecture, specific surface area, and acid–base properties of solid oxide catalysts. Consequently, the conversion of lignocellulosic biomass mediated by these catalysts necessitates a more nuanced understanding, accounting for the complexities arising from their unique physicochemical characteristics. Drawing from the aforementioned background, the present study sought to investigate the catalytic impacts of various inorganic acids/bases (including hydrochloric acid, sulfuric acid, nitric acid, sodium carbonate, and potassium carbonate) as well as nano-sized oxides (CeO_2, TiO_2, ZnO, Al_2O_3, MgO, SiO_2) on the hydrothermal liquefaction of cotton stalks under relatively mild conditions (220 °C for 4 h). The core focus of this study was to assess key indicators like the yield, composition, functional groups, calorific value, and energy recovery rate of the cotton stalk-derived bio-oil. Furthermore, the pore structure and acid–base properties of the nano-oxides were meticulously characterized to gain insights into their influence on the quality of the hydrothermal bio-oil. This research not only enhances our comprehension of the oxides' role in the hydrothermal conversion of lignocellulosic biomass but also holds significant implications for designing more efficient oxide-based catalysts and attaining high-yield bio-oil production from lignocellulosic materials at reduced temperatures.

2. Materials and Methods

2.1. Materials

Cotton stalks were collected from a farmland in Karamay, Xinjiang, China. The stalks were pulverized into a fine powder and then dried in an oven at 105 °C for 12 h under an air atmosphere, ready for use. Sulfuric acid (H_2SO_4), hydrochloric acid (HCl), nitric acid (HNO_3), sodium hydroxide (NaOH), and potassium hydroxide (KOH) were obtained from local suppliers. Dichloromethane (CH_2Cl_2), sodium carbonate (Na_2CO_3), potassium carbonate (K_2CO_3), nano-sized magnesium oxide (MgO, 50 nm), nano-sized zinc oxide (ZnO, 30 ± 10 nm), nano-sized cerium oxide (CeO_2, 20~50 nm), nano-sized titanium dioxide (TiO_2, rutile hydrophilic, 40 nm), nano-sized titanium dioxide (TiO_2, anatase hydrophilic, 40 nm), nano-sized silicon dioxide (SiO_2, 30 nm), and nano-sized aluminum oxide (Al_2O_3, α-phase mixture, 30 nm) were purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. (Shanghai, China). Ultra-pure water (DI water, 18.25 MΩ × cm) used in this study was prepared by a WPUP-UV-20 ultrapure water machine (Sichuan Water Technology Development Co., Ltd., Chengdu, China).

2.2. Hydrothermal Liquefaction Technology (HTL) Process and Product Separation Method

As illustrated in Figure 1, hydrothermal liquefaction experiments were carried out in a 100 mL high-temperature, high-pressure reactor constructed from 316 L stainless steel. The process was briefly outlined as follows: dried, powdered cotton stalks were combined with ultra-pure water at a solid-to-liquid ratio of 1:10, and a specific amount of nano-oxides (such as MgO, ZnO, CeO_2, TiO_2 (rutile), TiO_2 (anatase), SiO_2, Al_2O_3) or 1 mol/L acid/base catalysts (H_2SO_4, HCl, HNO_3, NaOH, KOH, Na_2CO_3, K_2CO_3) was added. The reactor was heated to 220 °C while stirring at 200 r/min and maintained for 4 h, after which it was cooled to room temperature using water. Gaseous products were vented and not collected, and the reactor was thoroughly rinsed with dichloromethane. The liquid and solid products were then transferred to a Büchner funnel for vacuum filtration. The solid product (biochar) was dried to a constant weight at 105 °C in an air atmosphere. The liquid product was further separated into aqueous and dichloromethane phases. The dichloromethane phase underwent reduced-pressure distillation at 40 °C to remove the solvent, resulting in the bio-oil (labeled as CS-HT-Oil, CS-HT-Oil-MgO, CS-HT-Oil-ZnO, CS-HT-Oil-CeO_2, CS-HT-Oil-TiO_2 (r), CS-HT-Oil-TiO_2 (a), CS-HT-Oil-SiO_2, CS-HT-Oil-Al_2O_3, CS-HT-Oil-H_2SO_4, CS-HT-Oil-HCl, CS-HT-Oil-HNO_3, CS-HT-Oil-NaOH, CS-HT-Oil-KOH, CS-HT-Oil-Na_2CO_3, CS-HT-Oil-K_2CO_3).

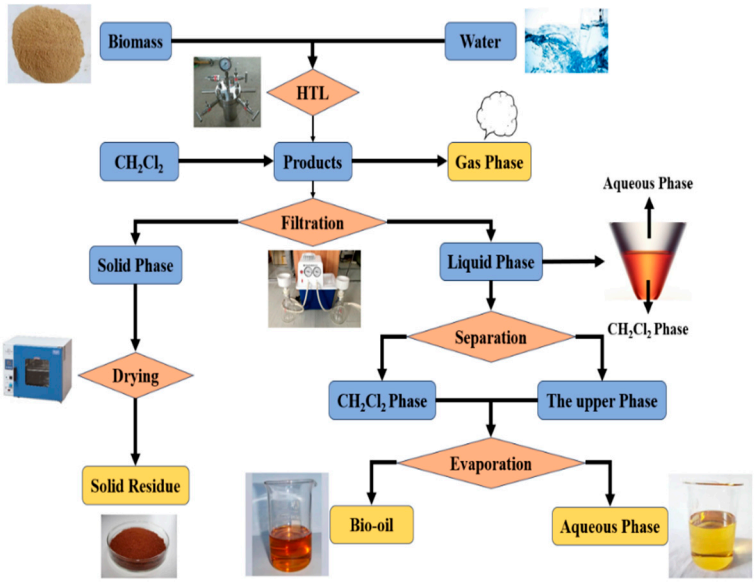

Figure 1. Separation process of hydrothermal liquefaction products.

The hydrothermal reactor diagram of the experimental device is shown in Figure 2.

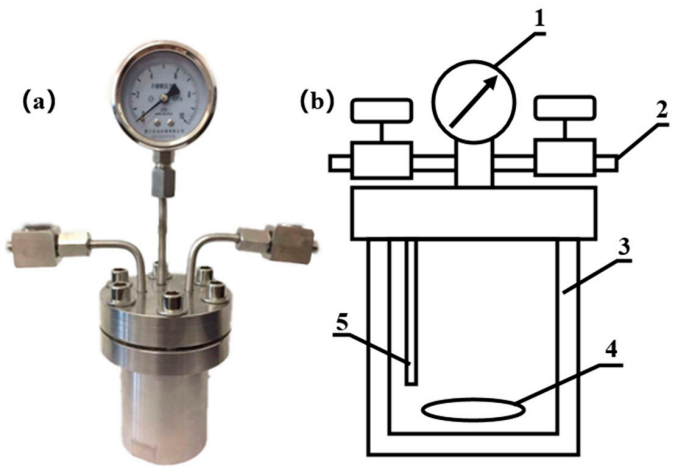

Figure 2. Hydrothermal reactor installation diagram. (**a**) Hydrothermal reactor physical picture. (**b**) Hydrothermal reactor perspective picture. 1: pressure gauge; 2: exhaust port; 3: polytetrafluoroethylene lining; 4: stirring magneton; 5: temperature sensor.

The calculation formulas for the yields of bio-oil, biochar, and other products are as follows:

$$\text{Bio-oil yield}(\%) = \frac{\text{mass of bio-oil}}{\text{mass of cotton stalk}} \times 100\% \quad (1)$$

$$\text{Solid yield}(\%) = \frac{\text{mass of solid}}{\text{mass of cotton stalk}} \times 100\% \quad (2)$$

$$\text{Others yield}(\%) = 100\% - \text{bio-oil yield} - \text{solid yield} \quad (3)$$

2.3. Materials and Product Analysis

The specific surface area and pore structure of the oxides were analyzed using a BEL-SORP-MIN adsorption instrument through nitrogen adsorption at 77 K. Prior to the measurements, the oxides were activated by heating at 100 °C for 6 h.

The acid–base properties of the oxides were assessed using temperature-programmed desorption (TPD) tests with NH_3 and CO_2 on a Catalyst Analyzer BELCAT-B equipped with a TCD detector.

The elemental analysis of the hydrothermal bio-oil was performed on an Elementar Vario EL cube elemental analyzer. The elemental contents of carbon (C), hydrogen (H), nitrogen (N), and sulfur (S) were determined.

The composition of the bio-oil was characterized using a gas chromatography–mass spectrometry (GC-MS) system (Agilent 7890B/5977).

Fourier transform infrared spectroscopy (FTIR) was conducted utilizing a Thermo Scientific iN10 spectrometer (Waltham, MA, USA), encompassing a wavenumber range of 500–4000 cm^{-1} for comprehensive spectral analysis, adopting the absorbance testing mode.

Thermogravimetric analysis (TGA) of the hydrothermal bio-oil was performed using a Netzsch STA409 integrated thermal analyzer, under a nitrogen atmosphere with a heating rate of 10 °C/min. The test temperature range was 30–800 °C.

3. Results

3.1. Effect of Catalytic on Products Distribution from HTL

The yield and product distribution of hydrothermal bio-oil catalyzed by nano-oxides are presented in Figure 3. Different nano-oxides have varying effects on the yield of catalyzed hydrothermal bio-oil. Ranked from highest to lowest, the yields are as follows: CeO_2 (7.96%), TiO_2 (rutile, 7.04%), ZnO (5.16%), Al_2O_3 (4.84%), MgO (4.35%), TiO_2 (anatase, 2.68%), and SiO_2 (1.27%). Compared to the yield of hydrothermal bio-oil without a catalyst (5.16%), the use of nano-TiO_2 (rutile) and nano-CeO_2 significantly enhances bio-oil production. In contrast, nano-TiO_2 (anatase) and nano-SiO_2 have inhibitory effects. Notably, CeO_2, a distinctive metal oxide, readily creates numerous oxygen vacancies due to its rapid interconversion between Ce^{3+} and Ce^{4+} oxidation states. These oxygen vacancies facilitate the activation of C-O bonds in the oxygen-containing groups within the biomass feedstock, enhancing the decarbonylation reactions during the hydrothermal liquefaction of cotton stalks. Consequently, the formation of additional hydrocarbons is promoted, ultimately leading to an increase in the yield of bio-oil [18].

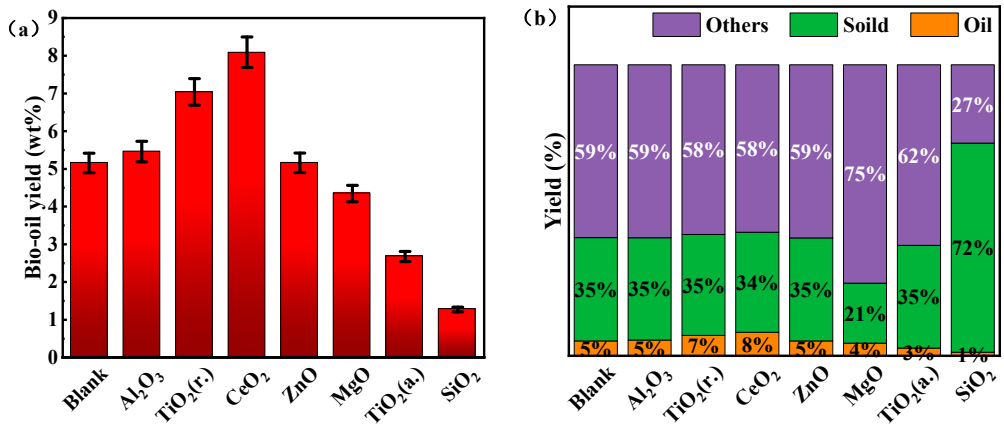

Figure 3. Yield of hydrothermal bio-oil in the presence of different oxides (**a**). Product distribution in the presence of different oxides (**b**).

From the perspective of cotton stalk hydrothermal bio-oil yield, to further ascertain the catalytic effect of CeO_2, various additions of CeO_2 were also investigated, as shown in Figure 4. The results show that as the amount of CeO_2 increases, the bio-oil yield initially increases and then decreases, although it remains higher than the yield from cotton stalk hydrothermal bio-oil without a catalyst. This indicates that CeO_2 effectively catalyzes the production of cotton stalk hydrothermal bio-oil. The observed initial increase and subsequent decrease in bio-oil yield with rising CeO_2 amounts suggest that both excessive

and insufficient quantities of CeO_2 are detrimental to bio-oil production when the biomass raw material amount is fixed.

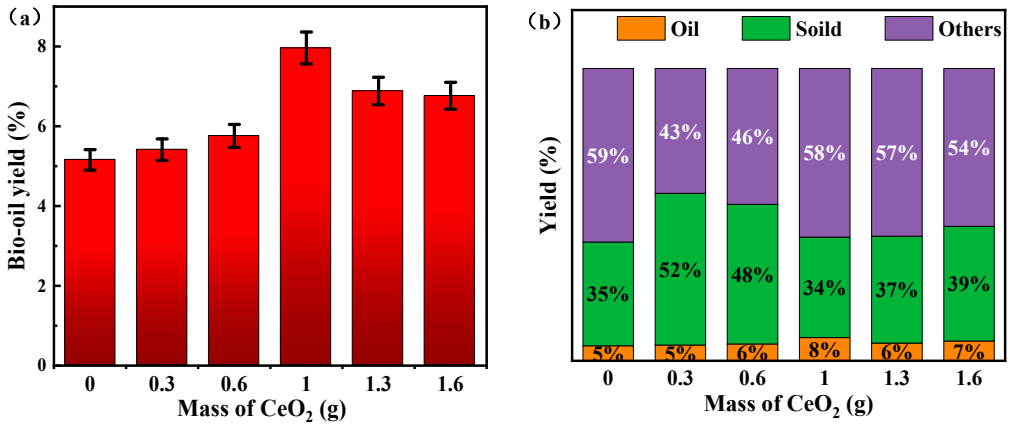

Figure 4. Yield of hydrothermal bio-oil with different CeO_2 dosages (**a**). Product distribution under different CeO_2 dosages (**b**).

The addition of CeO_2 demonstrated a peak effect, likely due to the crucial influence of the concentration of acid–base sites on its surface. To further elucidate the impact of acidity and alkalinity on the production of cotton stalk hydrothermal bio-oil, the catalytic effects of various inorganic acids and bases were also examined. The results of bio-oil yield and product composition are presented in Figure 5. Using the yield of hydrothermal bio-oil as an indicator, the catalytic activity of inorganic acids and bases for hydrothermal bio-oil can be ranked as follows: NaOH > HNO_3 > K_2CO_3 > Na_2CO_3 > KOH > HCl > H_2SO_4. Overall, the yields with acid and base catalysts were lower than those achieved with nano-oxide catalysts. Both types of catalysts inhibited bio-oil production compared to the yield obtained without any catalyst. Additionally, it was observed that different coordinating atoms had varying impacts on bio-oil production, with homogeneous bases generally exhibiting superior catalytic effects compared to homogeneous acids. Inorganic acids primarily act on the hemicellulose component of biomass raw materials, degrading it into monosaccharides and further converting them into bio-oil components [19]. However, the low content of hemicellulose in cotton stalks (only 15.64 wt% [20]) results in lower bio-oil yields with acid catalysis. The OH^- ions in alkaline catalysts can weaken the hydrogen bonds between cellulose and hemicellulose, saponify the ester bonds between hemicellulose and lignin, and break the ether bonds between hemicellulose and lignin. This effectively disrupts the connections among the three major components in straw, leading to an increased yield of bio-oil [21].

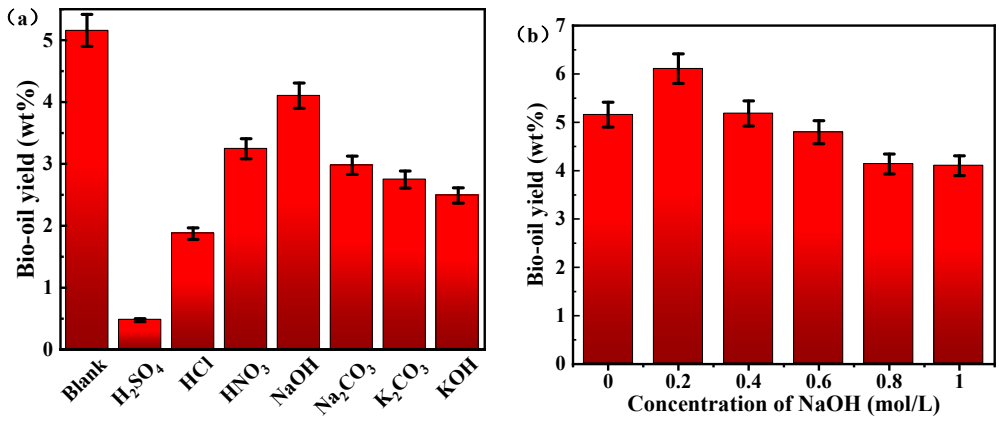

Figure 5. Yield of cotton stalk hydrothermal bio-oil catalyzed by different inorganic acids and bases (**a**). Yield of cotton stalk hydrothermal bio-oil catalyzed by different concentrations of NaOH (**b**).

From Figure 5a, it is evident that the hydrothermal bio-oil yield with HNO_3 as the catalyst is the highest, reaching 3.25%. In addition to serving as an inorganic acid catalyst, HNO_3 is also an oxidizing agent that can significantly disrupt the LCC structure in lignocellulosic biomass. As a result, at lower temperatures, it demonstrates superior catalytic performance compared to other acids. In contrast, alkali catalysts enhance bio-oil production by increasing the internal surface area of cellulose through swelling, reducing its crystallinity and degree of polymerization, and causing structural breakdown in lignin. These effects improve the reactivity of hemicellulose and cellulose even at lower temperatures. In alkaline solutions, metal ions enter the amorphous and crystalline regions of cellulose in hydrated and ionic forms. The smaller the radius of the alkali metal ion, the larger the diameter of its hydrated form and the more pronounced its swelling ability. Given that Na^+ has a smaller ionic radius compared to K^+, cotton stalk fibers treated with NaOH solution swell to a greater volume and exhibit decreased cohesion relative to those treated with KOH solution [22]. Therefore, the yield of cotton stalk hydrothermal oil catalyzed by K_2CO_3 is higher than that catalyzed by Na_2CO_3. The experimental results of this section indicate that the interaction of catalysts and other substances with the organizational structure of raw materials can also influence the yield of hydrothermal bio-oil during the hydrothermal conversion process [23].

As with CeO_2, an investigation was conducted to determine the optimal amount of NaOH required to maximize the yield of hydrothermal bio-oil. The results are presented in Figure 5b. Similar to CeO_2, the bio-oil yield decreased with increasing amounts of NaOH. Among the tested concentrations, 0.2 mol/L NaOH achieved the highest yield of 6.11%, surpassing the 5.16% yield obtained without a catalyst under identical conditions. This indicates that inorganic alkalis can indeed catalyze the hydrothermal process of cotton stalks, contributing to an increase in bio-oil yield. Combining the results from Figure 4a, we conclude that excessive addition of acids or alkalis is unfavorable from the perspective of the yield of cotton stalk hydrothermal bio-oil. For the production of cotton stalk hydrothermal bio-oil, there is an optimal ratio of surface-active catalytic components (such as functional groups or OH^- ions) within the catalyst. More detailed results will be discussed in Section 3.3.2.

It is noteworthy that the hydrothermal temperature utilized in this study was 220 °C, significantly lower than the typically used range of 280–350 °C reported in previous literature [21,24,25]. This finding highlights the significant reactivity of cotton stalks even at relatively low temperatures. By carefully selecting and designing catalysts with enhanced activity, it is possible to further optimize the yield of hydrothermal bio-oil at low temperatures. This represents a crucial step toward the industrialization of biomass feedstock conversion processes.

On the one hand, the hydrothermal method adopted in this study requires mild reaction conditions (220 °C, 4 h) for the preparation of bio-oil, which is far lower than the reaction temperature of 300–600 °C for the preparation of bio-oil by pyrolysis method [26,27]. At the same time, hydrothermal reaction can handle raw materials with high humidity and has low requirements for raw materials, while the pyrolysis method requires the pretreatment of raw materials to a low moisture content [28,29], which significantly increases the cost. On the other hand, this study systematically investigated the effects of the catalysts used on the bio-oiliness produced by hydrothermal reactions, especially the acidity–alkalinity of the catalysts, including the acidity–alkalinity of homogeneous catalysts and the acidity–alkalinity of heterogeneous catalysts, as well as the effect of the pore structure of the catalysts used on the bio-oiliness. This has a positive guiding effect on the transformation industry of biomass raw materials.

3.2. CS Bio-Oil Characterization

3.2.1. FTIR

The infrared spectrum FTIR analysis of different nanometer oxide catalyzed hydrothermal bio-oil is shown in Figure 6. The absorption peaks in the range of 3200–3500 cm^{-1}

belong to the stretching vibration of N-H and O-H, which come from the amine or amide, alcohol and phenol components in the bio-oil [30]. The asymmetric stretching vibration of C-H in the range of 2850 to 3000 cm^{-1} indicates that there may be alkane in the bio-oil [31]. The stretching vibration of carbonyl C=O in the range of 1850~1660 cm^{-1} indicates the presence of ketones, acids, lipids or anhydrides [32]. The stretching vibration of N-H in the range of 1660~1540 cm^{-1} indicates the presence of amide in the bio-oil [30]. The stretching vibration of carbon skeleton C=C in the range of 1500~1450 cm^{-1} is the stretching vibration of benzene ring, indicating the presence of aromatic compounds in the bio-oil [14]. The stretching vibration of methyl-CH_2 and methyl-CH_3 in the range of 1450~1470 cm^{-1} is the asymmetric deformation vibration, and the stretching vibration of methyl-CH_3 in the range of 1380~1370 cm^{-1} is the symmetric deformation vibration, indicating the presence of alkane in the bio-oil [31]. The stretching vibration of acyl in the range of 1220~1040 cm^{-1} is the stretching vibration of acyl, indicating the presence of amide compounds; the stretching vibration of ether group in the range of 1150~900 cm^{-1} is the stretching vibration of ether group, indicating the presence of ether compounds. The peak at 725 cm^{-1} indicates the presence of heterocyclic derivatives [1]. The infrared spectrum analysis results show that the bio-oil produced from cotton stalk through hydrothermal liquefaction contains aromatic compounds, organic acid compounds, ketone compounds, ester compounds, nitro-containing compounds, and acyl compounds. The rich functional groups indicate that bio-oil has important potential as a functional chemical [33].

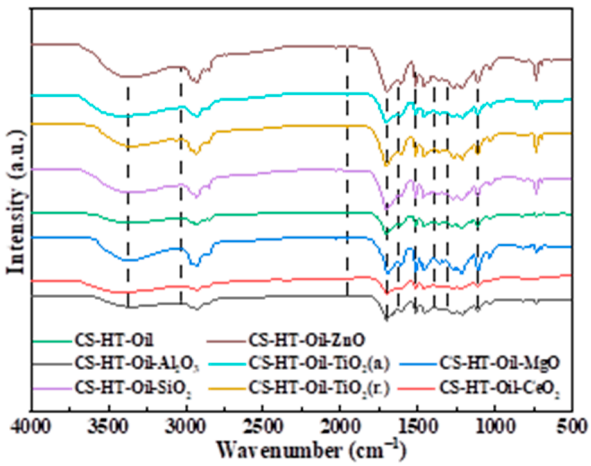

Figure 6. Infrared spectra of hydrothermal bio-oil catalyzed by nano oxides.

3.2.2. Elemental Analysis

The elemental analysis results of typical cotton stalk and hydrothermal bio-oil catalyzed by different nanometer oxides are shown in Table 1, where the calorific value calculation formula and energy recovery rate calculation formula are as follows [34]:

$$\mathrm{HHV(MJ/kg)} = 0.338 \times \mathrm{C} + 1.428 \times (\mathrm{H} - \mathrm{O}/8) + 0.095 \times \mathrm{S} \quad (4)$$

$$\text{Energy recovery}(\%) = \frac{\text{CS HHV} \times \text{Bio-oil yield}}{\text{CS HHV}} \times 100\% \quad (5)$$

Analysis of the results indicates that the addition of nanometer oxides significantly enhanced the hydrogen content and saturation level of the catalyzed hydrothermal bio-oil compared to that without a catalyst. The H/C ratio provides a reference for the content of aromatic compounds [35], and the increase in H/C ratio observed with the addition of nanometer oxides suggests that these oxides promote cracking and ring-opening reactions during the hydrothermal liquefaction process, thus reducing the content of aromatic compounds [36]. The O/C ratio is used to assess the level of hydrodeoxygenation in the bio-oil during hydrothermal liquefaction. The hydrothermal bio-oil catalyzed by nano-alumina

displayed relatively low carbon (C) and nitrogen (N) contents, but high sulfur (S) and oxygen (O) contents, resulting in a high O/C ratio. Since C=O bonds have lower energy contents than C-C bonds, the elevated oxygen content leads to a reduced calorific value, measured at 22.68 MJ/kg. In contrast, the hydrothermal bio-oil catalyzed by magnesia exhibited higher levels of C, hydrogen (H), and N, coupled with lower O content, yielding a lower O/C ratio. This indicates that nano-magnesia effectively facilitates the hydrodeoxygenation process, enhancing the calorific value of the bio-oil to 27.71 MJ/kg. As lignin is the carbon-richest component in cotton stalks, nano-magnesia may promote the decomposition of lignin, and phenolic oligomers obtained from lignin depolymerization reactions exist in the bio-oil components, leading to its high carbon content and low oxygen content [21]. Among the catalysts tested, silica-catalyzed hydrothermal bio-oil exhibited the lowest energy recovery rate of 1.85%, while ceria-catalyzed bio-oil showed the highest energy recovery rate of 11.27%, indicating its great potential as a high-quality biofuel [37].

Table 1. Elemental analysis of cotton stalk and hydrothermal bio-oil catalyzed by nano-oxides.

	C (wt%)	N (wt%)	H (wt%)	S (wt%)	O [1] (wt%)	H/C	O/C	HHV (MJ/kg)	Energy Recovery (%)
CS	45.96	1.02	5.97	0.09	43.5	1.56	0.71	16.30	-
CS-HT-Oil	61.52	0.62	5.734	0.262	31.864	1.12	0.39	23.32	7.38
CS-HT-Oil-SiO_2	61.00	0.69	6.086	0.342	31.882	1.2	0.39	23.65	1.85
CS-HT-Oil-TiO_2(r.)	61.46	0.76	5.978	0.445	31.357	1.17	0.38	23.76	10.26
CS-HT-Oil-CeO_2	60.25	1.00	5.905	0.513	32.332	1.18	0.40	23.07	11.27
CS-HT-Oil-Al_2O_3	59.41	0.75	5.822	1.319	32.699	1.18	0.41	22.68	6.73
CS-HT-Oil-ZnO	62.48	0.8	6.085	0.327	30.308	1.17	0.36	24.43	7.73
CS-HT-Oil-MgO	67.31	1.51	6.441	0.659	24.08	1.15	0.27	27.71	7.39

[1] O(wt%) = 100% − (C% + H% + S%).

3.2.3. TGA

TGA can provide insights into the combustion performance of bio-oil to a certain extent. As depicted in Figure 7, the TGA results of cotton stalk hydrothermal oil with nanometer oxides additions are presented.

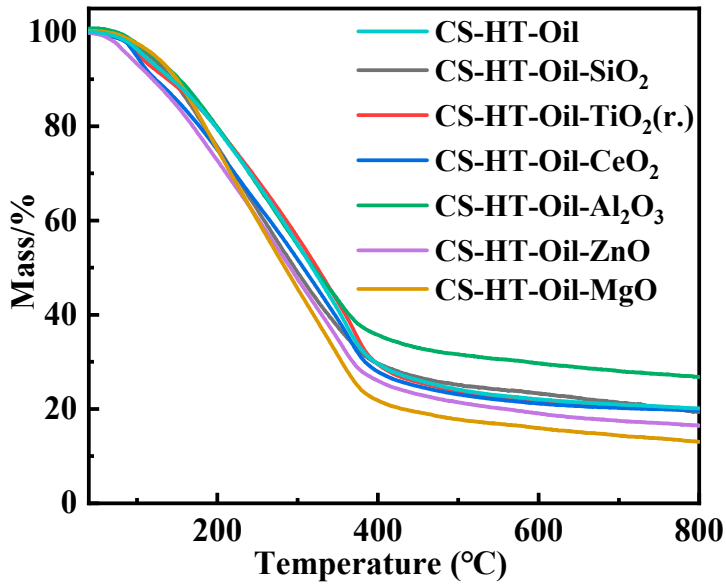

Figure 7. TGA of hydrothermal bio-oil catalyzed by nano oxides.

The results indicate that as the temperature rises, the weight loss curves of the bio-oil become steeper, but after reaching 380 °C, the trends of the curves flatten out signifi-

cantly. Compared to the cotton stalk hydrothermal bio-oil without any additives, all oxide-catalyzed cotton stalk hydrothermal oils exhibit higher ultimate rates within the temperature range of 0–800 °C. This suggests that the addition of oxides can enhance the combustion performance of cotton stalk hydrothermal bio-oil. Among them, MgO exhibits the most significant improvement. Below 220 °C, nano-zinc-oxide-catalyzed hydrothermal bio-oil has the fastest weight loss rate, with a remaining mass of 66 wt% at 220 °C and 23 wt% at 380 °C [3]. The TGA curves are primarily determined by the chemical composition and elemental composition of the bio-oil. Based on the elemental analysis results of cotton stalk hydrothermal bio-oil catalyzed by different nanometer oxides, CS-HT-Oil-MgO possesses the highest carbon and hydrogen contents and the lowest oxygen content, along with the maximum calorific value. Therefore, it is reasonable that CS-HT-Oil-MgO exhibits the best combustion performance [38].

3.2.4. GC-MS

The GC-MS was employed to analyze the hydrothermal bio-oil catalyzed by different nanometer oxides. The GC-MS results of the uncatalyzed hydrothermal bio-oil and the cotton stalk bio-oil catalyzed by CeO_2 with the highest yield are shown in Figures S1 and S2, and Tables S1 and S2. Compared to the uncatalyzed cotton stalk hydrothermal oil, which predominantly consists of nitrogen- and oxygen-containing heterocyclic compounds, phenols, and other aromatic substances, the bio-oil catalyzed by nano-CeO_2 exhibits a higher proportion of aliphatic compounds, including aldehydes, acids, aliphatic carbon ring compounds, and heterocyclic compounds. This indicates that during the hydrothermal process of cotton stalks, the oxygen vacancies generated by the transformation between Ce^{3+} and Ce^{4+} oxidation states of nano-CeO_2 facilitate the activation of C-O bonds, promoting reactions such as cleavage, decarboxylation, and ring-opening in the raw biomass feedstock, thereby enhancing the composition of the bio-oil [18].

A comprehensive comparison of the components of hydrothermal bio-oil catalyzed by various nanometer oxides, as shown in Table 2, reveals that the addition of nanometer oxides can promote the formation of compounds such as N-ethyl formamide and aliphatic aldehydes while reducing the production of nitrogen-containing aromatic substances.

Table 2. Comparison of hydrothermal bio-oil components catalyzed by nano-oxides (GC peak area%).

Entry	Compound	SiO_2	TiO_2	CeO_2	Al_2O_3	ZnO	MgO	Blank
1	N-ethyl formamide	19.83	8.43	27.62	9.42	13.47	-	-
2	5-Nitro-6-chloro-2,4(1H,3H)-Pyrimidinedione	-	10.12	-	5.74	-	-	-
3	Acetaldehyde	-	-	36.21	41.55		27.58	
4	1,1-Dimethylcyclopropane	-	-	19.91		34.07	-	-
5	2,4-Dichloro-5-oxo2-hexenedioic acid	-	-	8.23	19.91	19.93	-	-
6	2-Ethylacridine	-	-	-	-	-	1.75	11.22

Notably, the inclusion of nano-CeO_2 favors the generation of nitrogen-containing aliphatic compounds, aldehydes, acids, and cycloalkanes. These findings indicate that the addition of oxides can effectively modulate the composition of cotton stalk hydrothermal bio-oil, thereby enhancing both the yield and quality of the bio-oil to a certain extent [34].

3.3. Catalyst Characterization

3.3.1. BET

The BET characterization of the nanometer oxides used in the experiment is summarized in Table S3. Nano-SiO_2, which exhibited the lowest yield in catalyzing the hydrothermal bio-oil, the highest specific surface area of 189.03 m^2/g and the largest total

pore volume of 0.564 cm^3/g. Conversely, nano-CeO_2, which achieved the highest yield in catalyzing hydrothermal bio-oil, had the smallest total pore volume of 0.07885 cm^3/g and the narrowest average pore width of 10.04 nm. This suggests that a lower total pore volume is conducive to increasing the yield of bio-oil. A graphical representation of the correlation between the total pore volume and average pore width of the oxides and the energy recovery rate of cotton stalk hydrothermal bio-oil is presented in Figure 8. The results indicate that a smaller total pore volume and average pore diameter of the nanometer oxides are associated with a higher energy recovery rate of the catalyzed hydrothermal bio-oil. The biomass hydrothermal bio-oil is chiefly composed of decomposition products, or interaction products of these decomposition products, derived from the natural macromolecules in the biomass feedstock. The bio-oil energy recovery rate is primarily governed by the elemental composition and yield of the bio-oil, as determined by the calculation formula. A higher yield or calorific value of the bio-oil translates to a greater energy recovery rate. Additionally, a lower total pore volume and average pore diameter can significantly improve the bio-oil yield, thereby increasing the energy recovery rate.

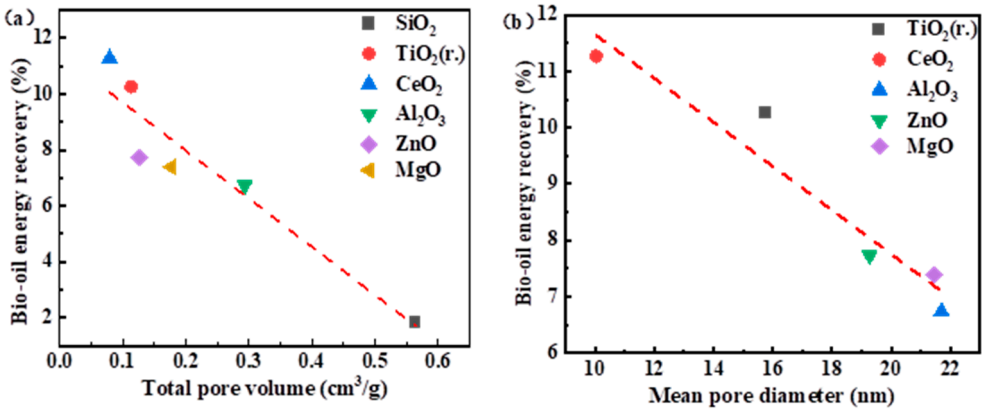

Figure 8. Comparison chart of the relationship between total pore volume and energy recovery rate of nano oxides (**a**). Comparison chart of the relationship between average pore width and energy recovery rate (**b**). (The dashed line is the guiding line).

3.3.2. TPD

As discussed in the section on the effects of varying CeO_2 and NaOH amounts on cotton stalk hydrothermal bio-oil yield, excessive functional group density and imbalanced acidity/alkalinity are detrimental to bio-oil production when the biomass feedstock is kept constant. In this section, NH_3-TPD and CO_2-TPD analyses were conducted on different nanometer oxides to measure their surface acid and base densities, as summarized in Table S4. Nano-SiO_2, which exhibited the lowest yield in catalyzing hydrothermal bio-oil, had the lowest acid value of 0.006 mmol/g. In contrast, nano-CeO_2, which achieved the highest yield, displayed the lowest base value of 0.010 mmol/g. Notably, despite having the highest acid and base values, nano-MgO yielded a relatively low amount of 4.35% bio-oil. This suggests that both excessively high and low acid values are detrimental to the production of catalyzed hydrothermal bio-oil, while a moderate base value is advantageous. Furthermore, nano-MgO, which produced bio-oil with the lowest oxygen content and highest calorific value, also exhibited the highest acid value of 0.15 mmol/g and base value of 0.098 mmol/g. These findings indicate that elevated acid and base values facilitate the hydrodeoxygenation process and enhance the calorific value of the catalyzed hydrothermal bio-oil. A graphical representation of the relationship between the acidic and basic sites of nanometer oxides and their energy recovery rates for cotton stalk hydrothermal bio-oil is presented in Figure 9. It reveals a nonlinear dependency between the acidic and basic sites of nanometer oxides and their energy recovery rates for catalyzed hydrothermal bio-oil. Specifically, a lower number of acidic and basic sites corresponds to a higher energy recovery rate for the catalyzed hydrothermal bio-oil.

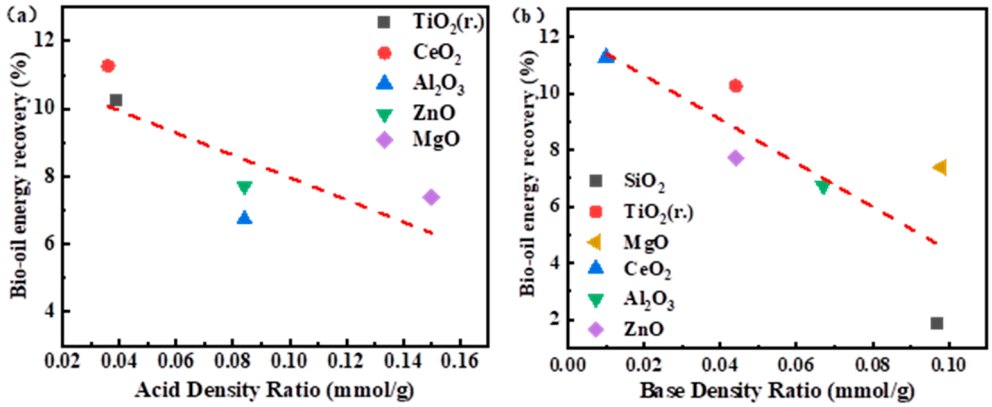

Figure 9. Comparison chart of the relationship between acidic sites and energy recovery rate of nano oxides (**a**). Comparison chart of the relationship between alkaline sites and energy recovery rate (**b**). (The dashed line is the guiding line).

By comparing the relationship between the ratio of base value to acid value of different nanometer oxides and the energy recovery rate of their catalyzed hydrothermal bio-oil (as shown in Figure 10), it can be observed that there is a "U-shaped" relationship between the base/acid value of nanometer oxides and the energy recovery rate of the catalyzed hydrothermal bio-oil. Both higher and lower base/acid ratios (around 1.1 and 0.3, respectively) are associated with higher energy recovery rates for hydrothermal bio-oil. This suggests that, for oxide-catalyzed cotton stalk hydrothermal bio-oil, the energy recovery rate results from the combined effects of acid and base sites. During the hydrothermal liquefaction of cotton stalks, the basic sites of nanometer oxides may help neutralize soluble acids, potentially mitigating autocatalysis caused by cracking. Meanwhile, their acid sites can promote depolymerization reactions, particularly targeting cellulose within lignocellulosic materials. Additionally, acid sites facilitate hydrolysis, dehydration, and decarboxylation reactions, which can lower the oxygen content in the hydrothermal bio-oil but may also reduce the yield of bio-oil to a certain extent [12].

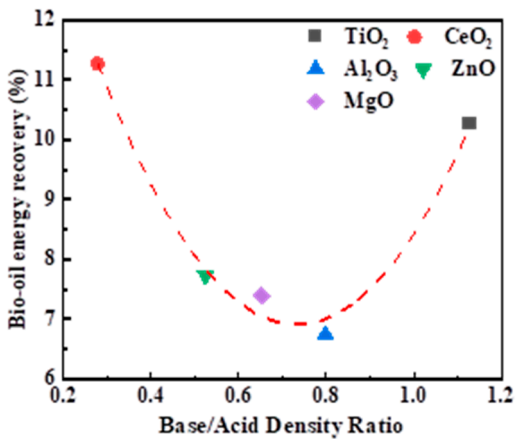

Figure 10. The relationship between the base value/acid value of nano oxides and their catalytic hydrothermal bio-oil energy recovery rate. (The dashed line is the guiding line).

By analyzing the signal intensities of a series of oxides through NH_3-TPD and CO_2-TPD at different temperature ranges, information on the strength of acid and base sites of the oxides can be obtained (shown in Figure S3). Nano-CeO_2, which achieves the highest yield in catalyzing hydrothermal bio-oil, has a relatively high number of weak base sites. Conversely, nano-SiO_2, which yields the lowest amount of bio-oil, possesses a higher quantity of weak acid sites. This suggests that a greater presence of weak base

sites enhances the yield of hydrothermal bio-oil, whereas an abundance of weak acid sites impedes its production. Notably, nano-MgO, which produces bio-oil with the lowest oxygen content and highest calorific value, exhibits the highest quantities of both medium acid and base sites. This suggests that a higher quantity of medium acid and base sites is conducive to the hydrodeoxygenation process and the enhancement of the calorific value of catalyzed hydrothermal bio-oil [4].

Currently, the preparation of lignocellulosic biomass predominantly relies on pyrolysis, which requires the raw material to undergo dewatering and demands significant energy input for high-temperature pyrolysis. In contrast, this study utilizes a relatively mild hydrothermal conversion method, which not only avoids the high energy consumption associated with raw material dewatering but also operates at lower temperatures, further reducing energy costs.

Moreover, most studies on the hydrothermal preparation of bio-oil from lignocellulosic biomass focus on the effects of operating parameters and catalysts on bio-oil yield and quality [39]. The results indicate that the acidity and alkalinity of the system are critical for bio-oil production. Therefore, in this study, we first employed homogeneous acid and base additives in the hydrothermal conversion of lignocellulosic biomass to directly investigate their impact on bio-oil yield. To address the challenge of separating the bio-oil from the catalyst, we also explored the effects of nanoscale oxides with distinct acidity, alkalinity, and porous structures as additives in the hydrothermal process. This approach aims to establish a correlation between the acid–base properties and structure of solid catalysts, and their influence on the yield and properties of hydrothermal bio-oil from lignocellulosic biomass. The findings from this research will provide insights for the design of more efficient and easily separable heterogeneous catalysts for hydrothermal biomass conversion.

At the same time, the applications of solid by-products, specifically biochar, are extensive. On one hand, biochar can be utilized for the adsorption of various heavy metals (such as cadmium, lead, and copper) and organic pollutants due to its large surface area and excellent adsorption properties, making it effective for water treatment and air purification. Additionally, biochar enhances soil moisture retention, improves soil structure, and can serve as a carrier for fertilizers, thereby increasing the organic matter content and fertility of the soil. On the other hand, it can be employed as a fuel source, contributing to renewable energy, and it can also be used as an additive in chemical manufacturing to produce other materials.

4. Conclusions

This comprehensive study delves into the intricate effects of various inorganic acids, bases, and nano-sized oxides on the mild hydrothermal liquefaction (220 °C, 4 h) of cotton stalks for the production of bio-oil. We systematically assessed the influence of these additives on crucial parameters, including bio-oil yield, functional group composition, elemental content, calorific value, and energy recovery rate. Notably, CeO_2 and NaOH emerged as potent enhancers, with 1 g of CeO_2 and 0.2 mol/L NaOH yielding the highest bio-oil outputs of 7.96 wt% and 6.11%, respectively. CeO_2's exceptional performance stems from its Ce^{3+}/Ce^{4+} redox cycle-induced oxygen vacancies, facilitating C-O bond activation, while NaOH's efficacy lies in its disruption of biomass hydrogen bonding, ester bond saponification, and Na+-mediated swelling, accelerating depolymerization. Analysis via infrared spectroscopy, elemental analysis, and thermogravimetry underscores the rich functional group content of oxide-catalyzed bio-oils, portending their potential as functional chemical precursors. Nano-oxides notably elevate bio-oil hydrogen content, enhancing its saturation and combustion performance. GC-MS profiling reveals nano-oxides' preference for N-ethyl formamide and aliphatic aldehyde formation while suppressing nitrogenous aromatics. Nano-CeO_2 specifically favors nitrogenous aliphatics, aldehydes, acids, and cycloalkanes, significantly upgrading the quality of bio-oil. Additionally, nano-oxide structural attributes correlate with energy recovery rates, with smaller pore volumes and widths, exemplified by nano-CeO_2 (0.07885 cm^3/g, 10.04 nm), yielding the highest, 11.27%,

recovery. Surface acid–base balance modulates bio-oil characteristics, with alkaline sites mitigating autocatalytic cracking and acidic sites driving depolymerization. Weak bases boost yield, while weak acids hinder it. Medium-strength acid–base sites facilitate hydrodeoxygenation, enhancing bio-oil calorific value. These findings illuminate the pivotal roles of oxides and inorganic acids/bases in lignocellulosic hydrothermal liquefaction, particularly agricultural and forestry waste conversion. They inform the rational design and synthesis of oxide-based catalysts for optimized bio-oil yield, quality, and energy recovery, fostering more efficient biomass conversion technologies.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/en17194854/s1, Figure S1: GC-MS of hydrothermal bio-oil without catalysts; Figure S2: GC-MS of hydrothermal bio-oil catalyzed by nano-sized CeO_2; Figure S3: NH_3-TPD (a) and CO_2-TPD (b) of nano oxides; Table S1: Components for catalyst-free hydrothermal bio-oil; Table S2: Components for CeO_2 catalyzed hydrothermal bio-oil; Table S3: Mesoporous structures of nano-oxides; Table S4: Acid and base densities of nano-oxides.

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